

Single-Asset Trading with Deterministic Alpha

gappy [+GPT 5.6 Sol Ultra]

Abstract

This note solves a finite-horizon trading problem for one asset with a known deterministic expected-return path. With quadratic trading costs, the optimal holding is the output of a linear two-sided exponential filter; a Dirichlet Green function gives the closed-form solution for a given alpha path. Replacing the quadratic cost by a 3/2-power cost produces an exact nonlinear costate boundary-value problem. For a general alpha path, that problem does not have an elementary closed-form solution. A formal perturbation about the quadratic model is available, but at power 3/2 it can be inaccurate. An iteratively reweighted quadratic scheme is both analytically transparent and numerically precise. In the example below its relative L^2 holding error is 9.0×10^{-8} against an independent shooting solution on the 4,001-point evaluation grid. The same analysis extends to several assets with a non-diagonal covariance matrix and generic positive vectors of asset-specific, diagonal trading costs: quadratic costs decouple in generalized risk-cost eigenmodes, whereas 3/2-power costs lead to a coupled nonlinear costate system.

1 Problem and conventions

Let $x(t)$ be the dollar holding at time $t \in [0, T]$, let $\alpha(t)$ be the known instantaneous expected return per dollar, and let σ be the constant percentage volatility. For expositional simplicity, assume α is continuously differentiable on $[0, T]$. The numerical event path below is continuous and differentiable on each side of its event time; the differential equations are applied separately on those two intervals. Let $\lambda > 0$ denote risk aversion and set

$$q := \lambda \sigma^2 > 0. \quad (1)$$

With holdings measured in dollars, λ has units of inverse dollars and q has units of inverse dollar-time. Thus q is the risk penalty per unit of squared dollar exposure. For a trading-cost power $p > 1$ and coefficient $a_p > 0$, let

$$C_p(v) := a_p |v|^p, \quad M_p(v) := \frac{dC_p(v)}{dv} \quad (2)$$

denote the trading-cost function and its marginal cost. The objective is

$$\max_{x \in W^{1,p}[0,T]} \mathcal{J}_p[x] := \int_0^T \left\{ \alpha(t)x(t) - qx(t)^2 - a_p |\dot{x}(t)|^p \right\} dt, \quad x(0) = x(T) = 0. \quad (3)$$

Thus the square in the quadratic specification applies to the trading rate, not to the entire integrand. The endpoint conditions represent entering and exiting the event with no position. Section 2.1 records the simple modification for free endpoints.

Standard coercivity arguments give existence, and the objective is strictly concave because expected return is linear while $qx^2 + a_p |\dot{x}|^p$ is strictly convex. Consequently, the first-order conditions identify the unique global maximizer, not merely a local trading rule.

We reserve $c := a_2$ for the quadratic model and $k := a_{3/2}$ for the 3/2-power model. They have different units, so setting them to the same number has no invariant meaning. In particular,

$$[c] = (\$/\text{time})^{-1}, \quad [k] = (\$/\text{time})^{-1/2}, \quad \left[\left(\frac{k}{c} \right)^2 \right] = \$/\text{time}. \quad (4)$$

Thus $(k/c)^2$ is a trading rate. Section 5.1 gives a path-based rule for mapping c to k .

2 Quadratic trading costs

Take $p = 2$. For a perturbation h satisfying $h(0) = h(T) = 0$, integration by parts gives

$$\frac{d}{d\epsilon} \mathcal{J}_2[x + \epsilon h] \Big|_{\epsilon=0} = \int_0^T \{ (\alpha - 2qx)h - 2c\dot{x}h \} dt \quad (5)$$

$$= \int_0^T \{ \alpha - 2qx + 2c\ddot{x} \} h dt. \quad (6)$$

The optimal holding therefore solves

$$-c\ddot{x}(t) + qx(t) = \frac{\alpha(t)}{2}, \quad x(0) = x(T) = 0. \quad (7)$$

Define the inverse smoothing length

$$\kappa := \sqrt{\frac{q}{c}} \quad (8)$$

and the Dirichlet Green function for $-d^2/dt^2 + \kappa^2$,

$$G_D(t, s) = \frac{\sinh(\kappa \min\{t, s\}) \sinh(\kappa [T - \max\{t, s\}])}{\kappa \sinh(\kappa T)}. \quad (9)$$

Proposition 2.1 (Closed-form quadratic strategy). *Under the standing assumptions, the unique solution of the quadratic-cost problem is*

$$x_2^*(t) = \frac{1}{2c} \int_0^T G_D(t, s) \alpha(s) ds. \quad (10)$$

Equation (10) makes the economics explicit. The investor averages alpha both before and after t , because changing today's holding affects future trading costs. Away from the endpoints, the equivalent infinite-horizon kernel is proportional to $\exp(-\kappa |t - s|)$. Quadratic trading costs therefore smooth alpha before applying the frictionless scaling $1/(2q)$; they do not simply shrink the position. The characteristic look-ahead and look-back horizon is $\sqrt{c/q}$. As $c \downarrow 0$, $x_2^* \rightarrow \alpha/(2q)$ in $L^2[0, T]$. Convergence is uniform on compact subintervals of $(0, T)$, while narrow boundary layers enforce the fixed endpoint holdings.

For the piecewise-exponential alpha used below, the integral in (10) can also be evaluated interval by interval. More directly, on an interval where $\alpha(t) = A_j e^{r_j t}$,

$$x_j(t) = B_j^+ e^{\kappa t} + B_j^- e^{-\kappa t} + \frac{A_j e^{r_j t}}{2c(\kappa^2 - r_j^2)}, \quad r_j^2 \neq \kappa^2. \quad (11)$$

In the resonant case $r_j = \pm \kappa$, the last term is replaced by $-A_j t e^{r_j t} / (4c r_j)$. The four constants are determined by $x(0) = x(T) = 0$ and continuity of x and $\dot{x}$ at the event time. Thus the example is literally elementary on each side of the event, while (10) is the cleaner closed form.

2.1 Free endpoints

If the initial and terminal holdings are not fixed, the boundary term in the first variation imposes $\dot{x}(0) = \dot{x}(T) = 0$. The formula in Proposition 2.1 remains valid after replacing G_D by the Neumann Green function

$$G_N(t, s) = \frac{\cosh(\kappa \min\{t, s\}) \cosh(\kappa [T - \max\{t, s\}])}{\kappa \sinh(\kappa T)}. \quad (12)$$

3 The 3/2-power trading cost

For a general $p > 1$, the Euler-Lagrange equation is

$$\alpha(t) - 2qx(t) + a_p p \frac{d}{dt} \left(\operatorname{sgn}(\dot{x}(t)) |\dot{x}(t)|^{p-1} \right) = 0. \quad (13)$$

At $p = 3/2$, introduce the marginal trading-cost variable

$$m(t) := \frac{3k}{2} \operatorname{sgn}(\dot{x}(t)) \sqrt{|\dot{x}(t)|}. \quad (14)$$

Thus m is the derivative of the trading-cost function $k|v|^{3/2}$ with respect to the trading rate v , evaluated at $v = \dot{x}$; it is the marginal cost of changing that rate. Here and below, $\operatorname{sgn}(0) := 0$. The transformation is invertible, including at zero trading speed, and turns (13) into the exact first-order system

$$\dot{x}(t) = \frac{4}{9k^2} m(t) |m(t)| \quad (15)$$

$$\dot{m}(t) = 2qx(t) - \alpha(t) \quad (16)$$

$$x(0) = 0 \quad (17)$$

$$x(T) = 0 \quad (18)$$

This is a nonlinear two-point boundary-value problem. Eliminating x gives the equivalent second-order boundary-value problem

$$\begin{aligned} \dot{m}(t) &= \frac{8q}{9k^2} m(t) |m(t)| - \alpha(t), \\ m(0) &= -\alpha(0), \quad m(T) = -\alpha(T). \end{aligned} \quad (19)$$

Indeed, $x = (\dot{m} + \alpha)/(2q)$, so the derivative conditions in (19) are exactly equivalent to the fixed endpoints for x . For a piecewise differentiable alpha, such as the event path below, the equation applies on each smooth interval and x and m remain continuous at the join. With free holdings at the two endpoints, the natural conditions are instead $m(0) = m(T) = 0$, equivalently $\dot{x}(0) = \dot{x}(T) = 0$. The nonlinear boundary-value problem has no elementary solution for a general forcing path α .

4 Approximations anchored to the quadratic solution

4.1 Formal perturbation in the cost exponent

To compare powers without hiding units, introduce a reference rate $v_\star$ and write

$$C_{2-\epsilon}(v) = cv_\star^2 \left(\frac{|v|}{v_\star} \right)^{2-\epsilon}. \quad (20)$$

This family starts from coefficient c at $\epsilon = 0$ and has $3/2$ -power coefficient $c\sqrt{v_\star}$ at $\epsilon = 1/2$. To connect a separately calibrated pair (c, k) , set

$$v_\star = \left(\frac{k}{c}\right)^2. \quad (21)$$

For small ϵ ,

$$C_{2-\epsilon}(v) = cv^2 \left[1 - \epsilon \log\left(\frac{|v|}{v_\star}\right) \right] + O(\epsilon^2). \quad (22)$$

This is a pointwise expansion in ϵ ; at $v = 0$ it is understood by the continuous convention $v^2 \log(|v|/v_\star) := 0$. Let $x_\epsilon = x_0 + \epsilon x_1 + O(\epsilon^2)$, where $x_0 = x_2^\star$, and put $v_0 = \dot{x}_0$. Define the continuous function

$$\psi_{v_\star}(v) := \begin{cases} v \left[2 \log\left(\frac{|v|}{v_\star}\right) + 1 \right], & v \neq 0, \\ 0, & v = 0. \end{cases} \quad (23)$$

Formally matching first-order terms in the Euler equation gives

$$2c\ddot{x}_1 - 2qx_1 = c \frac{d}{dt} \psi_{v_\star}(v_0), \quad x_1(0) = x_1(T) = 0. \quad (24)$$

Using (9) and integrating by parts avoids differentiating the continuous but generally nonsmooth function $\psi_{v_\star}(v_0)$:

$$x_1(t) = \frac{1}{2} \int_0^T \frac{\partial G_D(t, s)}{\partial s} \psi_{v_\star}(v_0(s)) ds. \quad (25)$$

The approximation for the requested power is $x_{3/2} \approx x_0 + \frac{1}{2}x_1$. It is useful when the exponent is close to two and trading speeds stay near $v_\star$. The numerical example shows that $\epsilon = 1/2$ is too large for this first-order formula to be relied upon.

4.2 Iteratively reweighted quadratic approximation

A much more accurate approximation preserves the convenience of a sequence of quadratic problems. Smooth the $3/2$ penalty $k|v|^{3/2}$ as

$$k\phi_\delta(v) := k(v^2 + \delta^2)^{3/4}. \quad (26)$$

Write $\mathcal{J}_{3/2, \delta}$ for the objective obtained by replacing $k|\dot{x}|^{3/2}$ with $k\phi_\delta(\dot{x})$. Since $z^{3/4}$ is concave for $z > 0$, its tangent gives the global bound

$$\phi_\delta(v) \leq \phi_\delta(v_r) + \frac{3}{4}(v_r^2 + \delta^2)^{-1/4}(v^2 - v_r^2). \quad (27)$$

Terms independent of v can be discarded. Given $v_r = \dot{x}_r$, define the local effective quadratic-cost coefficient

$$w_r(t) := \frac{3k}{4} (v_r(t)^2 + \delta^2)^{-1/4}. \quad (28)$$

The next iterate is the solution of the linear variable-coefficient problem

$$-\frac{d}{dt} (w_r(t) \dot{x}_{r+1}(t)) + qx_{r+1}(t) = \frac{\alpha(t)}{2}, \quad x_{r+1}(0) = x_{r+1}(T) = 0. \quad (29)$$

This is an iteratively reweighted least-squares (IRLS) scheme, equivalently an MM algorithm: majorize the cost and then minimize the surrogate. The bound in (27) touches at $v = v_r$, so an

exact surrogate solve does not decrease the fixed- δ smoothed objective. On a fixed time grid, the iterates converge to the unique smoothed maximizer x_δ^* when each surrogate is solved exactly. A continuation schedule should therefore solve each stage adequately before reducing δ .

There is also a direct bound on the smoothing error. Subadditivity of $z^{3/4}$ gives, for every feasible x ,

$$0 \leq \mathcal{J}_{3/2}[x] - \mathcal{J}_{3/2,\delta}[x] \leq Tk\delta^{3/2}. \quad (30)$$

If every smoothed problem is solved to convergence before δ is reduced, strong concavity from the risk term therefore yields

$$\|x_\delta^* - x_{3/2}^*\|_{L^2}^2 \leq \frac{Tk\delta^{3/2}}{q}. \quad (31)$$

If a smoothed stage is solved only to objective gap $\eta_\delta := \mathcal{J}_{3/2,\delta}[x_\delta^*] - \mathcal{J}_{3/2,\delta}[\hat{x}_\delta]$, the same argument bounds the computed iterate:

$$\|\hat{x}_\delta - x_{3/2}^*\|_{L^2}^2 \leq \frac{Tk\delta^{3/2} + \eta_\delta}{q}. \quad (32)$$

Thus both smoothing and optimization error must vanish along a continuation schedule. These bounds establish convergence but are too conservative to serve as practical stopping rules.

5 Numerical event-alpha example

Take

$$\alpha(t) = \begin{cases} \exp((t - \tau)/\tau), & t \leq \tau, \\ \exp(2(\tau - t)/\tau), & t > \tau, \end{cases} \quad (33)$$

$$T = 1, \quad \tau = \frac{2}{3}, \quad \sigma = 0.02, \quad c = 0.01.$$

The path is continuous at τ and differentiable on either side, with a kink at the event peak. The illustration sets $\lambda = 1$, so $q = 0.0004$. The coefficient $c = 0.01$ applies only to the quadratic trading cost.

5.1 Calibrating the 3/2 coefficient

There is no universal numerical conversion between c and k : the units differ by the square root of a trading rate. A practical closed-form rule matches the two marginal transaction costs along the quadratic benchmark. The notation below suppresses the rule's dependence on the fixed alpha path, q , T , and the endpoint conditions; it is not a conversion based on c alone. Let $v_c(t) := \dot{x}_{2,c}^*(t)$ and note that

$$M_2(v) = 2cv, \quad M_{3/2}(v) = \frac{3k}{2} \operatorname{sgn}(v) \sqrt{|v|}. \quad (34)$$

The least-squares marginal-cost match is

$$k_{\text{MC}}(c) := \arg \min_{k>0} \int_0^T [M_2(v_c(t)) - M_{3/2}(v_c(t))]^2 dt \quad (35)$$

$$= \frac{4c}{3} \frac{\int_0^T |v_c(t)|^{3/2} dt}{\int_0^T |v_c(t)| dt}. \quad (36)$$

This formula assumes $v_c \neq 0$. If the quadratic strategy never trades, the denominator vanishes and k is not identified by this matching criterion. Equivalently, $k_{\text{MC}} = (4c/3)\sqrt{v_{\text{eff}}}$, where $\sqrt{v_{\text{eff}}}$ is the turnover-weighted average of $\sqrt{|v_c|}$. This is the useful portable $c \mapsto k$ rule: solve the quadratic model once and evaluate two scalar integrals.

If the goal is specifically to make the holding paths as close as possible, choose a compact calibration interval $\mathcal{K} = [\underline{k}, \bar{k}] \subset (0, \infty)$ and define any path-optimal coefficient by

$$k_{L^2}(c) \in \operatorname{argmin}_{k \in \mathcal{K}} \int_0^T \left[x_{3/2,k}^*(t) - x_{2,c}^*(t) \right]^2 dt. \quad (37)$$

This criterion minimizes the absolute, rather than relative, L^2 distance. The reported relative discrepancy simply scales that distance by $\|x_{3/2,k}^*\|_{L^2}$ after calibration. On the compact interval $\mathcal{K}$, a minimizer exists because the optimal path varies continuously with k ; it need not be unique. For the example, the numerical search uses $\mathcal{K} = [k_{\text{MC}}/4, 4k_{\text{MC}}]$; a wider logarithmic sweep confirms that the reported minimizer is a unique interior minimum at the displayed numerical precision. For this example,

$$k_{\text{MC}}(0.01) = 0.0442185, \quad k_{L^2}(0.01) = 0.0470029. \quad (38)$$

Thus the closed-form rule is within six percent of the direct path-optimal calibration and is an effective initial value for the scalar search. The numerical comparison below uses $k = 0.0470029$, which is path-optimal on the stated calibration interval. For the exponent perturbation, equation (21) then gives $v_* = 22.0927$.

The exact $p = 3/2$ benchmark was obtained by shooting system (15) with an endpoint residual below 2×10^{-8} . The reweighted quadratic method started from the exact $p = 2$ solution and used $\delta = 10^{-1}, 10^{-2}, \dots, 10^{-6}$, stopping each stage when the relative L^2 iterate change was below 10^{-10} . In evaluating (25), the integral is split at $s = t$ because $\partial_s G_D(t, s)$ has a jump of magnitude one there; applying one quadrature rule across that jump would introduce a first-order mesh error. Continuous peak times are recovered from the zero of trading speed, equivalently from the zero of the marginal trading-cost variable. For a candidate path $\tilde{x}$, the reported relative L^2 error is

$$\frac{\|\tilde{x} - x_{3/2}^*\|_{L^2}}{\|x_{3/2}^*\|_{L^2}}, \quad (39)$$

evaluated by Simpson quadrature on the $N = 4,001$ grid.

Table 1: Strategies evaluated under the $p = 3/2$ objective. Holding errors are relative to the independent shooting solution. Objectives and error norms use $N = 4,001$ mesh quadrature. The quadratic peak time is analytic; the remaining peak times are interpolated continuous zero crossings of trading speed.

Method	Peak x	Peak time	$\mathcal{J}_{3/2}$	Relative L^2 error
Quadratic-cost rule	4.47069	0.53126	0.643572	0.07490
First-order perturbation	3.96687	0.58685	0.677436	0.06037
Reweighted quadratic	4.09673	0.54084	0.684290	9.00×10^{-8}
Exact $p = 3/2$ shooting	4.09673	0.54084	0.684290	0

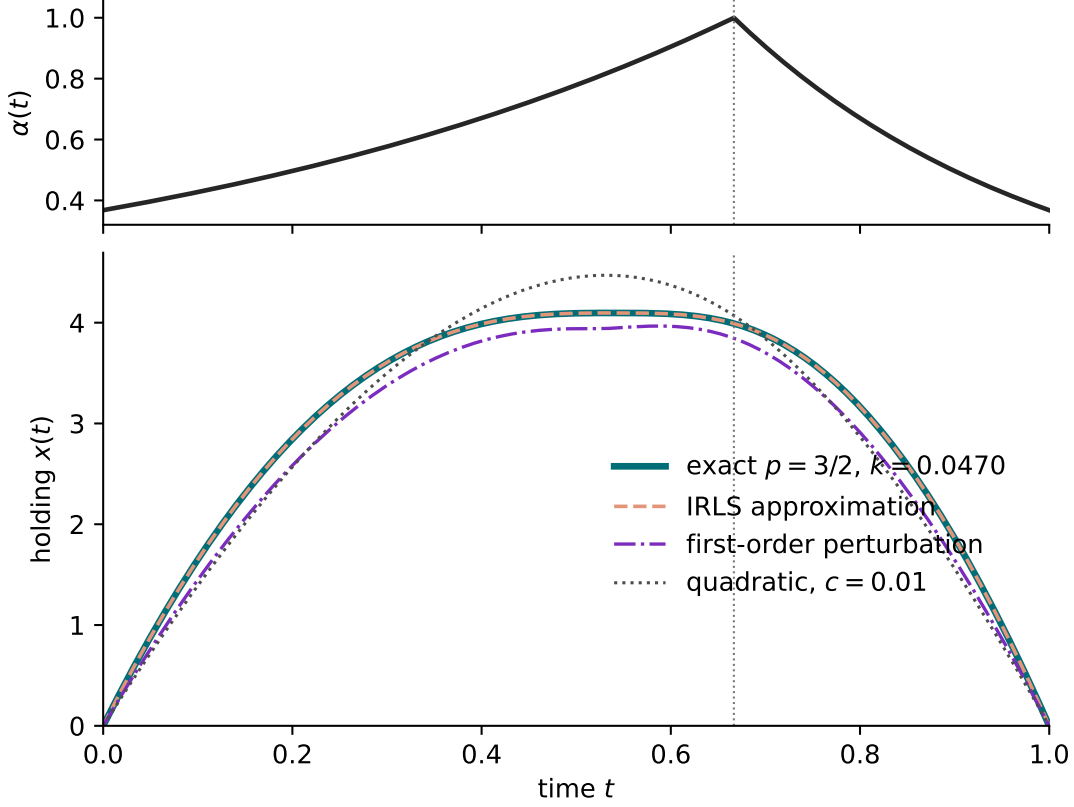


Figure 1: The deterministic alpha and calibrated optimal holdings. The quadratic curve uses $c = 0.01$, the exact $3/2$ curve uses the path-optimal $k = 0.0470029$, and the reweighted path is visually indistinguishable from the exact nonlinear solution.

Three conclusions matter in implementation. First, the asymmetric alpha decay makes the nonlinear position peak at $t = 0.54084$, well before the alpha peak at $\tau = 2/3$; the investor starts unwinding early because post-event alpha decays twice as fast as pre-event alpha rises. Second, coefficient matching matters: using the same number for c and k creates an economically meaningless scale change, whereas (37) gives an L^2 path discrepancy equal to 7.49% of $\|x_{3/2}^*\|_{L^2}$. A constant coefficient cannot make the paths identical because the two marginal costs have different speed dependence. Third, the calibrated exponent perturbation reduces the holding error further to 6.04%, while the reweighted quadratic approximation closes essentially the entire remaining gap: its relative sup-norm holding error is 1.35×10^{-7} and its objective gap is 1.73×10^{-7} , or 2.52×10^{-7} of the optimum, on the $N = 4,001$ evaluation grid. These last figures are mesh diagnostics rather than continuum error bounds; increasing the grid to $N = 16,001$ reduces the relative L^2 discrepancy to 5.61×10^{-9} .

For reference, under the quadratic objective the exact quadratic strategy has $\mathcal{J}_2 = 1.020721$. Under the $3/2$ objective, the exact solution decomposes approximately into gross alpha 2.048801, risk penalty 0.004069, and trading cost 1.360442, leaving $\mathcal{J}_{3/2} = 0.684290$.

6 Multivariate extension

Let $\mathbf{x}(t), \boldsymbol{\alpha}(t) \in \mathbb{R}^n$ be the vectors of dollar holdings and known alpha curves. Assume that every component of $\boldsymbol{\alpha}$ is continuously differentiable on $[0, T]$. Let $\boldsymbol{\Sigma} \in \mathbb{R}^{n \times n}$ be the symmetric positive-definite instantaneous covariance matrix of per-dollar returns and define the risk-penalty matrix

$$\mathbf{Q} := \lambda \boldsymbol{\Sigma}. \quad (40)$$

Thus $\mathbf{x}' \boldsymbol{\Sigma} \mathbf{x}$ is the instantaneous variance rate of dollar profit and loss. The alpha paths may be estimated independently, but once they are treated as known deterministic functions their probabilistic independence is not needed. A non-diagonal $\boldsymbol{\Sigma}$ couples the optimal holdings through $\mathbf{Q}$. Fix arbitrary positive cost vectors

$$\mathbf{c} := (c_1, \dots, c_n)' \in \mathbb{R}_{++}^n, \quad \mathbf{k} := (k_1, \dots, k_n)' \in \mathbb{R}_{++}^n, \quad (41)$$

and define the diagonal coefficient matrices

$$\mathbf{C} := \text{diag}(\mathbf{c}), \quad \mathbf{K} := \text{diag}(\mathbf{k}). \quad (42)$$

Thus transaction costs remain separable across assets; heterogeneity enters only through the diagonal entries. The equal-cost assumptions $\mathbf{C} = c\mathbf{I}$ and $\mathbf{K} = k\mathbf{I}$ are special cases.

6.1 Quadratic costs: exact modal solution

The multivariate quadratic problem is

$$\max_{\mathbf{x}} \int_0^T \{ \boldsymbol{\alpha}(t)' \mathbf{x}(t) - \mathbf{x}(t)' \mathbf{Q} \mathbf{x}(t) - \dot{\mathbf{x}}(t)' \mathbf{C} \dot{\mathbf{x}}(t) \} dt, \quad \mathbf{x}(0) = \mathbf{x}(T) = \mathbf{0}. \quad (43)$$

For a vector perturbation $\mathbf{h}$ that vanishes at both endpoints, the first variation is

$$\delta \mathcal{J}_2 = \int_0^T \mathbf{h}(t)' \{ \boldsymbol{\alpha}(t) - 2\mathbf{Q} \mathbf{x}(t) + 2\mathbf{C} \dot{\mathbf{x}}(t) \} dt. \quad (44)$$

Hence the unique optimum solves the linear matrix boundary-value problem

$$-\mathbf{C} \ddot{\mathbf{x}}(t) + \mathbf{Q} \mathbf{x}(t) = \frac{\boldsymbol{\alpha}(t)}{2}, \quad \mathbf{x}(0) = \mathbf{x}(T) = \mathbf{0}. \quad (45)$$

Because $\mathbf{Q}$ and $\mathbf{C}$ are positive definite, there is a $\mathbf{C}$ -orthonormal basis of generalized eigenvectors $\{\mathbf{v}_j\}_{j=1}^n$ satisfying

$$\mathbf{Q} \mathbf{v}_j = \kappa_j^2 \mathbf{C} \mathbf{v}_j, \quad \mathbf{v}_i' \mathbf{C} \mathbf{v}_j = \mathbf{1}_{\{i=j\}}. \quad (46)$$

These modes are explicit: if

$$\mathbf{C}^{-1/2} \mathbf{Q} \mathbf{C}^{-1/2} = \mathbf{U} \text{diag}(\kappa_1^2, \dots, \kappa_n^2) \mathbf{U}', \quad \mathbf{U}' \mathbf{U} = \mathbf{I}, \quad (47)$$

then $\mathbf{v}_j = \mathbf{C}^{-1/2} \mathbf{u}_j$, where $\mathbf{u}_j$ is column j of $\mathbf{U}$. Hence arbitrary positive c_i require only a diagonal whitening before the ordinary symmetric eigendecomposition. Write $\mathbf{x}(t) = \sum_j \mathbf{v}_j z_j(t)$ and left-multiply (45) by $\mathbf{v}_j'$. Each modal holding then solves

$$-\ddot{z}_j(t) + \kappa_j^2 z_j(t) = \frac{\mathbf{v}_j' \boldsymbol{\alpha}(t)}{2}, \quad z_j(0) = z_j(T) = 0. \quad (48)$$

Define

$$G_{j,D}(t, s) := \frac{\sinh(\kappa_j \min\{t, s\}) \sinh(\kappa_j [T - \max\{t, s\}])}{\kappa_j \sinh(\kappa_j T)}. \quad (49)$$

The exact multivariate strategy is therefore

$$\mathbf{x}_2^*(t) = \frac{1}{2} \sum_{j=1}^n \mathbf{v}_j \int_0^T G_{j,D}(t, s) \mathbf{v}_j' \boldsymbol{\alpha}(s) ds. \quad (50)$$

Every generalized risk-cost mode is thus a two-sided exponential filter with characteristic horizon $1/\kappa_j$, where κ_j^2 is the generalized eigenvalue of the pair $(\mathbf{Q}, \mathbf{C})$. With heterogeneous costs it is not, in general, an eigenvalue of $\boldsymbol{\Sigma}$ alone, and a mode that mixes assets has no single asset-specific trading-cost coefficient. The alpha of one asset generally changes the holding of every other asset through the generalized eigenvectors $\mathbf{v}_j$.

For the common-cost special case $\mathbf{C} = c\mathbf{I}$, write $\boldsymbol{\Sigma} = \mathbf{U} \text{diag}(s_1, \dots, s_n) \mathbf{U}'$ with $\mathbf{U}'\mathbf{U} = \mathbf{I}$. Then $\kappa_j = \sqrt{\lambda s_j/c}$ and (50) becomes

$$\mathbf{x}_2^*(t) = \frac{1}{2c} \mathbf{U} \int_0^T \text{diag}(G_{1,D}(t, s), \dots, G_{n,D}(t, s)) \mathbf{U}' \boldsymbol{\alpha}(s) ds. \quad (51)$$

For $n = 1$, this reduces exactly to (10). With free endpoints, replace every $G_{j,D}$ by its Neumann counterpart.

6.2 Separable 3/2 costs: exact coupled system

With the generic positive vector $\mathbf{k}$ defined above, the natural componentwise nonlinear problem is

$$\max_{\mathbf{x}} \int_0^T \left\{ \boldsymbol{\alpha}' \mathbf{x} - \mathbf{x}' \mathbf{Q} \mathbf{x} - \sum_{i=1}^n k_i |\dot{x}_i|^{3/2} \right\} dt, \quad \mathbf{x}(0) = \mathbf{x}(T) = \mathbf{0}. \quad (52)$$

Define the marginal trading-cost variables component by component,

$$m_i(t) := \frac{3k_i}{2} \text{sgn}(\dot{x}_i(t)) \sqrt{|\dot{x}_i(t)|}. \quad (53)$$

The inverse transformation is $\dot{x}_i = 4m_i |m_i| / (9k_i^2)$. Consequently, the exact optimality system is

$$\dot{\mathbf{x}}(t) = \frac{4}{9} \mathbf{K}^{-2} (m_i(t) |m_i(t)|)_{i=1}^n, \quad (54)$$

$$\dot{\mathbf{m}}(t) = 2\mathbf{Q}\mathbf{x}(t) - \boldsymbol{\alpha}(t), \quad (55)$$

$$\mathbf{x}(0) = \mathbf{0}, \quad (56)$$

$$\mathbf{x}(T) = \mathbf{0}. \quad (57)$$

The off-diagonal elements of $\mathbf{Q} = \lambda \boldsymbol{\Sigma}$ enter the second equation and couple all assets. Eliminating $\mathbf{x}$ gives the equivalent second-order boundary-value problem

$$\begin{aligned} \ddot{\mathbf{m}}(t) &= \frac{8}{9} \mathbf{Q} \mathbf{K}^{-2} (m_i(t) |m_i(t)|)_{i=1}^n - \dot{\boldsymbol{\alpha}}(t), \\ \dot{\mathbf{m}}(0) &= -\boldsymbol{\alpha}(0), \quad \dot{\mathbf{m}}(T) = -\boldsymbol{\alpha}(T). \end{aligned} \quad (58)$$

The boundary conditions follow from $\mathbf{x} = \frac{1}{2}\mathbf{Q}^{-1}(\mathbf{m} + \boldsymbol{\alpha})$. If all endpoint holdings are free, the conditions are $\mathbf{m}(0) = \mathbf{m}(T) = \mathbf{0}$; the same condition applies componentwise to any asset whose endpoint holding is free.

Unlike the quadratic case, transformation to the generalized risk-cost eigenbasis does not decouple the nonlinear system: the componentwise 3/2 penalty is not invariant under a nontrivial rotation. An elementary closed form is therefore unavailable in general. If $\mathbf{Q}$ is diagonal in the original asset basis, the system separates into n copies of the single-asset problem.

6.3 Componentwise coefficient matching

The scalar $c \mapsto k$ rule also extends component by component. Let $\mathbf{v}_c(t) := \dot{\mathbf{x}}_{2,c}^*(t)$ be the trading-rate vector from the coupled quadratic problem with coefficient vector $\mathbf{c}$. Assume first that every component has nonzero reference turnover. Minimizing the aggregate squared discrepancy between quadratic and 3/2 marginal costs gives

$$\mathbf{k}_{\text{MC}}(\mathbf{c}) := \arg \min_{\mathbf{k} \in \mathbb{R}_{++}^n} \sum_{i=1}^n \int_0^T \left[2c_i v_{c,i}(t) - \frac{3k_i}{2} \text{sgn}(v_{c,i}(t)) \sqrt{|v_{c,i}(t)|} \right]^2 dt, \quad (59)$$

$$k_{\text{MC},i}(\mathbf{c}) = \frac{4c_i \int_0^T |v_{c,i}(t)|^{3/2} dt}{3 \int_0^T |v_{c,i}(t)| dt}, \quad i = 1, \dots, n. \quad (60)$$

If $v_{c,i} \equiv 0$, its k_i is not identified by marginal-cost matching and the corresponding ratio in (59) is left undefined. Although the formula is componentwise, each reference path $v_{c,i}$ depends on the full covariance matrix and the entire vector $\mathbf{c}$. Thus covariance coupling is respected while the calibration reduces to n independent scalar regressions after the quadratic solution is known. If exact holding-path matching is desired, choose a compact positive calibration box

$$\mathcal{K} := \prod_{i=1}^n [k_i, \bar{k}_i] \subset \mathbb{R}_{++}^n. \quad (61)$$

The direct analogue of (37) is then

$$\mathbf{k}_{L^2}(\mathbf{c}) \in \arg \min_{\mathbf{k} \in \mathcal{K}} \int_0^T \left\| \mathbf{x}_{3/2,\mathbf{k}}^*(t) - \mathbf{x}_{2,\mathbf{c}}^*(t) \right\|_2^2 dt. \quad (62)$$

A minimizer exists on the compact box $\mathcal{K}$ because the optimal path varies continuously with $\mathbf{k}$, although covariance coupling can prevent uniqueness. Formula (59) provides a natural initialization for this finite-dimensional search.

6.4 Multivariate reweighted-quadratic approximation

For a smoothing vector $\boldsymbol{\delta} = (\delta_1, \dots, \delta_n) \in \mathbb{R}_{++}^n$, let $\mathcal{J}_{3/2,\boldsymbol{\delta}}$ replace each penalty $k_i |\dot{x}_i|^{3/2}$ by $k_i (\dot{x}_i^2 + \delta_i^2)^{3/4}$. At iteration r , define

$$\mathbf{W}_r(t) := \text{diag} \left(\frac{3k_i}{4} [\dot{x}_{i,r}(t)^2 + \delta_i^2]^{-1/4} \right)_{i=1}^n. \quad (63)$$

The next iterate is obtained from the linear, risk-coupled problem

$$-\frac{d}{dt} (\mathbf{W}_r(t) \dot{\mathbf{x}}_{r+1}(t)) + \mathbf{Q} \mathbf{x}_{r+1}(t) = \frac{\boldsymbol{\alpha}(t)}{2}, \quad \mathbf{x}_{r+1}(0) = \mathbf{x}_{r+1}(T) = \mathbf{0}. \quad (64)$$

This is the multivariate analogue of (29). A standard time discretization gives a symmetric, positive-definite block system that is tridiagonal in time. At fixed δ , an exact MM update does not decrease the smoothed objective. On a fixed finite grid, the iterates converge to the unique smoothed maximizer $\mathbf{x}_\delta^*$ when each surrogate is solved exactly.

Define $E_\delta := T \sum_i k_i \delta_i^{3/2}$. The componentwise version of (30) gives

$$0 \leq \mathcal{J}_{3/2}[\mathbf{x}] - \mathcal{J}_{3/2,\delta}[\mathbf{x}] \leq E_\delta. \quad (65)$$

Hence an approximate smoothed solution $\hat{\mathbf{x}}_\delta$ satisfies the following bound, where $\eta_\delta \geq 0$ denotes the difference between the optimal and achieved values of the smoothed objective:

$$\|\hat{\mathbf{x}}_\delta - \mathbf{x}_{3/2}^*\|_{L^2}^2 \leq \frac{E_\delta + \eta_\delta}{\lambda_{\min}(\mathbf{Q})}. \quad (66)$$

Thus a continuation schedule converges in $L^2([0, T]; \mathbb{R}^n)$ when each stage is solved so that $E_\delta + \eta_\delta \rightarrow 0$.

Section 7 gives the finite-difference calculation used in practice. It preserves the divergence form in (64), so no derivative of a time-varying weight is formed explicitly.

6.5 First-order exponent approximation

The scalar exponent perturbation extends to correlated assets. Correlation does not alter the componentwise expansion of the trading costs; it enters when the resulting forcing is propagated through the common quadratic operator $-\mathbf{C}d^2/dt^2 + \mathbf{Q}$. Consequently, the first-order approximation requires only one additional solve with the same risk-cost-coupled operator used for the quadratic strategy.

For reference trading rates $v_{\star,i} > 0$, introduce the componentwise cost homotopy

$$\mathcal{E}_{i,2-\epsilon}(v_i) := c_i v_{\star,i}^2 \left(\frac{|v_i|}{v_{\star,i}} \right)^{2-\epsilon}, \quad i = 1, \dots, n. \quad (67)$$

At $\epsilon = 0$ this gives $c_i v_i^2$, while at $\epsilon = 1/2$ it gives $c_i \sqrt{v_{\star,i}} |v_i|^{3/2}$. Thus prescribed quadratic and 3/2 coefficients are connected by

$$v_{\star,i} = \left(\frac{k_i}{c_i} \right)^2, \quad i = 1, \dots, n. \quad (68)$$

This componentwise identity matches units as well as coefficients: each $v_{\star,i}$ is a trading rate.

For fixed v_i , expansion about $\epsilon = 0$ gives

$$\mathcal{E}_{i,2-\epsilon}(v_i) = c_i v_i^2 \left[1 - \epsilon \log \left(\frac{|v_i|}{v_{\star,i}} \right) \right] + O(\epsilon^2). \quad (69)$$

This expansion is pointwise in ϵ ; at $v_i = 0$ use the continuous convention $v_i^2 \log(|v_i|/v_{\star,i}) := 0$. Define

$$\psi_i(v) := \begin{cases} v \left[2 \log \left(\frac{|v|}{v_{\star,i}} \right) + 1 \right], & v \neq 0, \\ 0, & v = 0, \end{cases} \quad \boldsymbol{\psi}(\mathbf{v}) := (\psi_i(v_i))_{i=1}^n. \quad (70)$$

Then

$$\frac{\partial \mathcal{E}_{i,2-\epsilon}}{\partial v_i}(v_i) = 2c_i v_i - \epsilon c_i \psi_i(v_i) + O(\epsilon^2). \quad (71)$$

Write the optimizer along the homotopy as

$$\mathbf{x}_{2-\epsilon} = \mathbf{x}_0 + \epsilon \mathbf{x}_1 + O(\epsilon^2), \quad \mathbf{v}_0 := \dot{\mathbf{x}}_0, \quad (72)$$

where $\mathbf{x}_0 = \mathbf{x}_2^*$ solves (45). Substitution into the Euler equation and matching the terms linear in ϵ gives, in the weak or distributional sense,

$$-\mathbf{C}\ddot{\mathbf{x}}_1 + \mathbf{Q}\mathbf{x}_1 = -\frac{1}{2}\mathbf{C}\frac{d}{dt}\boldsymbol{\psi}(\mathbf{v}_0), \quad (73)$$

$$\mathbf{x}_1(0) = \mathbf{x}_1(T) = \mathbf{0}. \quad (74)$$

This equation is linear even though its forcing is constructed from the quadratic trading-rate path. Because $\mathbf{Q}$ need not be diagonal, a forcing generated by one asset generally changes the correction for every asset.

It is unnecessary to differentiate $\boldsymbol{\psi}(\mathbf{v}_0)$ numerically. For every test path $\mathbf{h} \in H_0^1((0, T); \mathbb{R}^n)$, the equivalent weak form is

$$\int_0^T \{\dot{\mathbf{h}}' \mathbf{C} \dot{\mathbf{x}}_1 + \mathbf{h}' \mathbf{Q} \mathbf{x}_1\} dt = \frac{1}{2} \int_0^T \dot{\mathbf{h}}' \mathbf{C} \boldsymbol{\psi}(\mathbf{v}_0) dt. \quad (75)$$

Alternatively, use the generalized eigenvectors and Green functions from (46). Integration by parts in the Green representation gives the derivative-free formula

$$\mathbf{x}_1(t) = \frac{1}{2} \sum_{j=1}^n \mathbf{v}_j \int_0^T \frac{\partial G_{j,D}(t, s)}{\partial s} \mathbf{v}_j' \mathbf{C} \boldsymbol{\psi}(\mathbf{v}_0(s)) ds, \quad (76)$$

$$\hat{\mathbf{x}}_{3/2}^{(1)}(t) := \mathbf{x}_0(t) + \frac{1}{2} \mathbf{x}_1(t) \quad (77)$$

$$= \mathbf{x}_0(t) + \frac{1}{4} \sum_{j=1}^n \mathbf{v}_j \int_0^T \frac{\partial G_{j,D}(t, s)}{\partial s} \mathbf{v}_j' \mathbf{C} \boldsymbol{\psi}(\mathbf{v}_0(s)) ds. \quad (78)$$

The factor 1/2 in the last approximation is the target exponent change $\epsilon = 2 - 3/2$. When $n = 1$, these expressions reduce to (24)–(25); when $\mathbf{Q}$ is diagonal in the asset basis, they separate asset by asset.

In a discretization, first solve for $\mathbf{x}_0$, evaluate $\boldsymbol{\psi}(\dot{\mathbf{x}}_0)$ on the time grid, and solve (75) for $\mathbf{x}_1$. The quadratic system matrix and its factorization can be reused. If the modal integral is evaluated directly, split the quadrature at $s = t$, where $\partial G_{j,D}(t, s)/\partial s$ has a jump. The approximation is formal: it is asymptotic as $\epsilon \rightarrow 0$, whereas the desired value is $\epsilon = 1/2$. It is most informative when the economically important trading rates are not many orders of magnitude from their reference rates. For production accuracy at power 3/2, the continuation IRLS scheme above is more reliable; the first-order strategy is useful as an inexpensive diagnostic or initializer.

Remark 6.1 (Portfolio-norm transaction cost). If the intended nonlinear cost is instead $k \|\dot{\mathbf{x}}\|_2^{3/2}$, the marginal trading-cost vector is the continuous map

$$\mathbf{m} := \begin{cases} \frac{3k}{2} \|\dot{\mathbf{x}}\|_2^{-1/2} \dot{\mathbf{x}}, & \dot{\mathbf{x}} \neq \mathbf{0}, \\ \mathbf{0}, & \dot{\mathbf{x}} = \mathbf{0}. \end{cases} \quad (79)$$

The first equation in (54) is then replaced by

$$\dot{\mathbf{x}} = \frac{4}{9k^2} \|\mathbf{m}\|_2 \mathbf{m}, \quad (80)$$

while the risk costate equation is unchanged.

7 Implementation recipe and conclusion

For quadratic costs, use the Green-function formula (10), or solve the tridiagonal discretization of (7). For the 3/2 cost, a robust production procedure is:

1. initialize with the quadratic strategy;
2. form the local coefficients in (28);
3. solve the linear boundary-value problem (29);
4. repeat to convergence, reduce δ , and finally verify the result by substituting into the exact costate system (15).

For the multivariate separable 3/2 cost, use the following grid calculation:

1. **Set inputs and grid.** Choose $t_j = jh$, $j = 0, \dots, N$, with $h = T/N$; evaluate $\alpha_j := \alpha(t_j)$; and supply $\mathbf{Q} \succ 0$, $\mathbf{k} \in \mathbb{R}_{++}^n$, a continuation factor $0 < \rho_\delta < 1$, and numerical tolerances. Set $\mathbf{x}_0 = \mathbf{x}_N = \mathbf{0}$.
2. **Initialize.** Use the coupled quadratic strategy for $\{\mathbf{x}_j^{(0)}\}_{j=0}^N$. Compute its edge rates by the formula in Step 3 and choose, for example,

$$\delta_i^{(0)} = \max \left\{ \gamma \max_j |v_{i,j+1/2}^{(0)}|, \delta_{i,\text{floor}} \right\}, \quad \gamma > 0.$$

Warm-start every later smoothing level from the preceding solution.

3. **Compute edge rates and IRLS weights.** At inner iteration r and smoothing level ℓ , calculate

$$\begin{aligned} \mathbf{v}_{j+1/2}^{(r)} &:= \frac{\mathbf{x}_{j+1}^{(r)} - \mathbf{x}_j^{(r)}}{h}, \\ \mathbf{W}_{j+1/2}^{(r)} &:= \text{diag} \left(\frac{3k_i}{4} \left[(v_{i,j+1/2}^{(r)})^2 + (\delta_i^{(\ell)})^2 \right]^{-1/4} \right)_{i=1}^n, \quad j = 0, \dots, N-1. \end{aligned}$$

4. **Solve the block-tridiagonal system.** Stack the interior holdings in $\mathbf{X}^{(r+1)}$ and solve

$$\mathbf{A}^{(r)} \mathbf{X}^{(r+1)} = \mathbf{b}.$$

Its nonzero blocks are

$$\begin{aligned} \mathbf{A}_{j,j}^{(r)} &= \mathbf{Q} + \frac{\mathbf{W}_{j-1/2}^{(r)} + \mathbf{W}_{j+1/2}^{(r)}}{h^2}, & \mathbf{b}_j &= \frac{\alpha_j}{2}, \\ \mathbf{A}_{j,j-1}^{(r)} &= -\frac{\mathbf{W}_{j-1/2}^{(r)}}{h^2}, & \mathbf{A}_{j,j+1}^{(r)} &= -\frac{\mathbf{W}_{j+1/2}^{(r)}}{h^2}. \end{aligned}$$

for $j = 1, \dots, N-1$. The matrix is symmetric positive definite, so use sparse Cholesky or a block-tridiagonal solver.

5. **Converge each smoothing level and continue.** Repeat Steps 3–4 until both the relative Frobenius change in the grid path and the smoothed stationarity residual are below tolerance; also confirm that the smoothed objective has not decreased. Set $\delta^{(\ell+1)} = \rho_\delta \delta^{(\ell)}$ and repeat. For a target L^2 holding error ε_x , continue until

$$\frac{T \sum_{i=1}^n k_i (\delta_i^{(\ell)})^{3/2}}{\lambda_{\min}(\mathbf{Q})} \leq \varepsilon_x^2$$

and make the inner-solve error negligible on the same scale.

6. **Validate the nonsmoothed solution.** From the final edge rates form

$$\mathbf{m}_{j+1/2} := \left(\frac{3k_i}{2} \operatorname{sgn}(v_{i,j+1/2}) \sqrt{|v_{i,j+1/2}|} \right)_{i=1}^n,$$

$$\mathbf{r}_j := \frac{\mathbf{m}_{j+1/2} - \mathbf{m}_{j-1/2}}{h} - 2\mathbf{Q}\mathbf{x}_j + \boldsymbol{\alpha}_j.$$

Require the interior residuals to be small. Repeat the calculation on a finer grid until the holdings are stable, then return $\{\mathbf{x}_j\}_{j=0}^N$.

The quadratic-cost model is fully explicit for any deterministic alpha. The 3/2 model has an exact costate characterization, but the forced nonlinear equation (19) prevents a general elementary solution. The exponent perturbation is analytically informative but not sufficiently accurate at the requested calibration. Iteratively reweighted quadratic solves retain the tractability of the quadratic model and, in this example, reproduce the true nonlinear solution to substantially better than one part in a million. With several assets, the quadratic model remains explicit after rotating into the generalized eigenmodes of $(\mathbf{Q}, \mathbf{C})$ for any $\mathbf{c} \in \mathbb{R}_{++}^n$. The separable 3/2 cost allows any $\mathbf{k} \in \mathbb{R}_{++}^n$ and does not share this modal decoupling, but its coupled costate system (54), componentwise calibration (59), first-order quadratic correction (77), and block-tridiagonal IRLS update (64) provide exact and practical counterparts to the single-asset results.